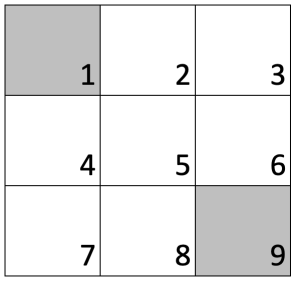
Exercise A



Let's consider the generic Markov Decision Process (MPD) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and a generic policy π.

1. Provide the pseudo-code for the Iterative Policy Evaluation algorithm to estimate V_π in the generic MDP; in the algorithm consider
   - having reached K iterations as unique exit condition;
   - don't report the 'in-place' version of the algorithm.
2. Consider the deterministic small gridworld reported in the Figure above: the action space is $\mathcal{A} = \{\text{up}, \text{down}, \text{left}, \text{right}\}$ and if an action takes the agent off the grid, the state remain unchanged. The state space is $\mathcal{S} = \{1, \ldots, 9\}$ and the terminal states are reported in grey. All actions are associated, no matter the state they are taken from, with a reward equals to -1. The state transitions are the obvious ones. Consider a random policy π, ie:

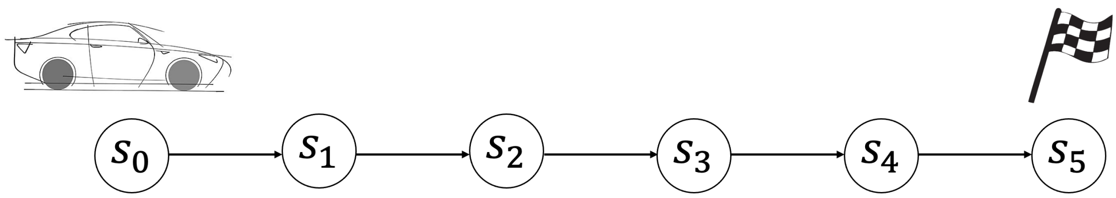
$$\pi(\text{up}, s_j) = \pi(\text{down}, s_j) = 0.5 \qquad \forall s_j \in \mathcal{S}$$

$$\pi(\text{left}, s_j) = \pi(\text{right}, s_j) = 0 \qquad \forall s_j \in \mathcal{S}$$

   Estimate V_π by using iterative policy evaluation with:
   - $\gamma = 1$;
   - initial estimation $V_\pi = (0, 0, 0, 0)$;
   - don't consider the 'in-place' version of iterative policy evaluation;
   - stop the algorithm after reaching $K = 4$ as exit condition.

   Report the procedure with calculations.
3. Suppose to apply Policy Improvement after k steps of the policy evaluation procedure described above. Report the obtained policy after $k = 1, \ldots, 4$ steps. Is it possible to derive an optimal policy? After how many steps?

Exercise B



Let's consider the generic Markov Decision Process (MPD) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ and a generic policy π.

1. Provide the pseudo-code for the TD(0) algorithm to estimate V_π in the generic MDP; in the algorithm consider
   - the incremental estimation for quantities evaluation based on the step size α.
2. Consider the 'driving home' example represented in the figure above: the goal of the agent is to estimate the remaining time before reaching the terminal state s_5 ('home'). The action space is $\mathcal{A}$ = drive and the state space is $\mathcal{S} = \{s_0, \ldots, s_5\}$. The state transitions are as depicted in the figure. Report the estimations for $V(s)$ with the Monte Carlo algorithm for policy evaluation considering:
   - a initial estimation $V(s_0) = 40$, $V(s_1) = 30$, $V(s_2) = 20$, $V(s_3) = 10$, $V(s_4) = 5$, $V(s_5) = 0$;
   - an episode where the collected rewards were (r_{t+1} is the notation for the reward associated with the transition from s_t and s_{t+1}): $r_1 = 15$, $r_2 = 8$, $r_3 = 12$, $r_4 = 6$, $r_5 = 4$;
   - $\alpha = 0.5$ and $\gamma = 1$.
3. Given the same settings described at the previous point, report the estimations for $V(s)$ with the TD(0) algorithm for policy evaluation.

Question 1
Let's consider the k-Armed Bandit problem.

1. Let's define the greedy policy: $\mathcal{A}_t =?, Q_t(a) =?$.
2. Let's consider the action a' and be Q_n its action value estimation after n times the action a' was taken. Define an efficient way to compute on-line Q_n.
3. Let's define the ϵ-greedy policy: $\mathcal{A}_t =?$. How are the not greedy actions chosen?
4. Let's define the Gradient Bandit Algorithm: $Pr\{\mathcal{A}_t\} =?, H_t(\cdot) =?$. How are the not greedy actions chosen?
5. List other approaches for dealing with the exploration-exploitation dilemma in k-armed bandits.

Question 2

Address the following questions.

- State the Policy Gradient Theorem for the episodic case.
- Show how the Policy Gradient Theorem can be used to develop the REINFORCE algorithm.

Question 3

Describe the backward-view T$D(\lambda)$ algorithm. Show that for $\lambda = 0$ the backward-view T$D(\lambda)$ algorithm is equivalent to the T$D(0)$ algorithm.

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